AN IMAGE OF A SPECTRUM THAT IS STABLE UNDER SPECIALIZATION IS CLOSED

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Points of the closure of a subset via nonempty basic opens. Used in the first proof (lines 9845-9848). Characterises membership in the closure as nonvanishing after inverting every element outside the prime.
- Minimal primes of the image of an injective ring map. Used in the second proof (lines 9861-9864). Shows that the image contains all minimal primes, which combined with specialization stability gives closedness.

Source reference. Original lemma 00FK: sources/stacks-algebra/pinned-original-source.tex, source lines 6123–6123. Navigation pointer for \ref{lemma-injective-minimal-primes-in-image} used inside the counted span (line 9863); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $R \to S$ be a ring map. Let $T \subset \mathrm{Spec}(R)$ be the image of $\mathrm{Spec}(S)$. If T is stable under specialization, then T is closed.*

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Proof. We give two proofs.

First proof. Let $\mathfrak{p} \subset R$ be a prime ideal such that the corresponding point of $\mathrm{Spec}(R)$ is in the closure of T. This means that for every $f \in R$, $f \notin \mathfrak{p}$ we have $D(f) \cap T \neq \emptyset$. Note that $D(f) \cap T$ is the image of $\mathrm{Spec}(S_f)$ in $\mathrm{Spec}(R)$. Hence we conclude that $S_f \neq 0$. In other words, $1 \neq 0$ in the ring S_f. Since $S_\mathfrak{p}$ is the directed colimit of the rings S_f we conclude that $1 \neq 0$ in $S_\mathfrak{p}$. In other words, $S_\mathfrak{p} \neq 0$ and considering the image of $\mathrm{Spec}(S_\mathfrak{p}) \to \mathrm{Spec}(S) \to \mathrm{Spec}(R)$ we see there exists a $\mathfrak{p}' \in T$ with $\mathfrak{p}' \subset \mathfrak{p}$. As we assumed T closed under specialization we conclude $\mathfrak{p}$ is a point of T as desired.

Second proof. Let $I = \mathrm{Ker}(R \to S)$. We may replace R by R/I. In this case the ring map $R \to S$ is injective. By Lemma 00FK all the minimal primes of R are contained in the image T. Hence if T is stable under specialization then it contains all primes. $\square$